DIGITAL PRINCIPLES AND COMPUTER ORGANIZATION

Question Bank with Multiple-Choice Options and Answers

Course Code: U25ADG01 **Regulation:** R2025
Institution: KPR Institute of Engineering and Technology
Coverage: Units I–V | Part A – 1 Mark MCQs

Prepared from the uploaded 2026–2027 Pattern A question bank. Questions and options follow the source; concise answer explanations are added for revision.

Note: Unit V, Part A, Question 14 has no readable question stem in the uploaded source, so it is not reconstructed or guessed.

UNIT I – COMBINATIONAL LOGIC

PART A – 1 MARK QUESTIONS

1. A combinational circuit is one whose output depends on:

A) Present inputs only

B) Previous inputs only

C) Clock pulse only

D) Memory elements only

Correct answer: A) Present inputs only
A combinational circuit has no memory; its output depends only on current inputs.

2. The number of cells in a 3-variable K-map is:

A) 4

B) 6

C) 8

D) 16

Correct answer: C) 8
A K-map has 2^n cells; for 3 variables, 2^3 = 8.

3. The minterm represented by binary 101 is:

A) m4

B) m5

C) m6

D) m7

Correct answer: B) m5
Binary 101 is decimal 5, so the minterm is m5.

4. For a half adder, A=1 and B=1. The Sum output is:

A) 0

B) 1

C) 2

D) Undefined

Correct answer: A) 0
Sum = A XOR B = 1 XOR 1 = 0.

5. A minterm contains:

A) All variables

B) Some variables

C) Only complemented variables

D) Only uncomplemented variables

Correct answer: A) All variables
Each minterm contains every variable exactly once, complemented or uncomplemented.

6. A decoder converts:

A) Binary information to unique outputs

B) Analog to digital

C) Many outputs to one input

D) Parallel to serial

Correct answer: A) Binary information to unique outputs
A decoder activates one output corresponding to the binary input.

7. The output of a multiplexer is selected by:

A) Data lines

B) Select lines

C) Enable line

D) Clock signal

Correct answer: B) Select lines
Select lines determine which data input reaches the output.

8. For a full adder, A=1, B=0 and Cin=1. The Sum output is:

A) 0

B) 1

C) 2

D) 3

Correct answer: A) 0
Sum = A XOR B XOR Cin = 1 XOR 0 XOR 1 = 0.

9. A multiplexer is also called a:

A) Data selector

B) Data converter

C) Data encoder

D) Data comparator

Correct answer: A) Data selector
A multiplexer selects one of several inputs.

10. A demultiplexer has:

A) One input and many outputs

B) Many inputs and one output

C) One input and one output

D) Many inputs and many outputs

Correct answer: A) One input and many outputs
A demultiplexer routes one input to one selected output.

11. The output line activated in a 3-to-8 decoder for input 101 is:

A) Y3

B) Y4

C) Y5

D) Y6

Correct answer: C) Y5
Input 101 is decimal 5, so output Y5 is selected.

12. The borrow output of a half subtractor for A=0 and B=1 is:

A) 0

B) 1

C) A

D) B

Correct answer: B) 1
Borrow = A'B = 1.

13. A priority encoder gives priority to:

A) Highest-order active input

B) Lowest-order input

C) Clock input

D) Enable input

Correct answer: A) Highest-order active input
A priority encoder resolves multiple active inputs by assigning priority, commonly to the highest-order input.

14. A magnitude comparator is used to:

A) Compare two binary numbers

B) Store data

C) Count pulses

D) Generate parity

Correct answer: A) Compare two binary numbers
It determines whether one binary number is greater than, equal to, or less than another.

15. The simplified form of A + AB is:

A) AB

B) B

C) A

D) None of the above

Correct answer: C) A
By the absorption law, A + AB = A.

16. A 4-to-16 decoder requires how many input lines?

A) 2

B) 3

C) 4

D) 16

Correct answer: C) 4
A decoder with 16 outputs needs log2(16) = 4 inputs.

17. A 3-to-8 decoder has:

A) 3 outputs

B) 5 outputs

C) 8 outputs

D) 6 outputs

Correct answer: C) 8 outputs
A 3-to-8 decoder has eight output lines.

18. Adjacent cells in a K-map differ by:

A) One variable

B) Two variables

C) Three variables

D) Four variables

Correct answer: A) One variable
Adjacent Gray-code cells differ in exactly one variable.

19. The simplified form of AB + AB' is:

A) A

B) B

C) AB

D) 1

Correct answer: A) A
AB + AB' = A(B+B') = A.

20. For a full adder with A=1, B=1, Cin=0, Carry equals:

A) 0

B) 1

C) 2

D) 3

Correct answer: B) 1
Two input ones produce a carry-out of 1.

UNIT II – SYNCHRONOUS SEQUENTIAL LOGIC

PART A – 1 MARK QUESTIONS

1. A negative edge-triggered flip-flop responds at:

A) 0→1 transition

B) 1→0 transition

C) Logic HIGH

D) Logic LOW

Correct answer: B) 1→0 transition
Negative-edge triggering occurs on the falling edge.

2. A T flip-flop toggles when:

A) T=0

B) T=1

C) Clock=0

D) Reset=1

Correct answer: B) T=1
T=1 causes the output to complement at the active clock edge.

3. A T flip-flop has T=1 continuously. Initially Q=1. Output after five clock pulses is:

A) 0

B) 1

C) Unknown

D) Invalid

Correct answer: A) 0
The output toggles five times; an odd number of toggles changes 1 to 0.

4. The JK flip-flop eliminates the _____ condition of S-R flip-flop:

A) Hold

B) Set

C) Invalid

D) Reset

Correct answer: C) Invalid
JK operation defines the condition J=K=1 as toggle, avoiding the invalid SR condition.

5. Which distinguishes a sequential circuit from a combinational circuit?

A) Uses only logic gates

B) Output depends on present inputs and previous states

C) Does not require memory

D) Output depends only on present inputs

Correct answer: B) Output depends on present inputs and previous states
Sequential circuits include memory and depend on current inputs and stored state.

6. In a Mealy machine, the output depends on:

A) Present state only

B) Input only

C) Present state and input

D) Next state only

Correct answer: C) Present state and input
Mealy outputs are functions of present state and present input.

7. The excitation inputs required to change a JK flip-flop from 0 to 1 are:

A) J=0, K=X

B) J=X, K=0

C) J=1, K=X

D) J=X, K=1

Correct answer: C) J=1, K=X
To set Q from 0 to 1, J must be 1; K is don't-care.

8. A 4-bit binary counter can count from:

A) 0 to 7

B) 0 to 15

C) 1 to 16

D) 0 to 16

Correct answer: B) 0 to 15
A 4-bit counter has 2^4 = 16 states, from 0 through 15.

9. Which memory element is level-triggered?

A) Master-slave flip-flop

B) Latch

C) JK flip-flop

D) Edge-triggered flip-flop

Correct answer: B) Latch
A latch responds while its enable level is active.

10. A master-slave flip-flop is triggered:

A) When clock is HIGH

B) When clock is LOW

C) When clock changes state

D) When a pulse is applied at clock input

Correct answer: C) When clock changes state
The master and slave stages operate on opposite clock levels, producing edge-like operation.

11. State minimization helps in reducing the:

A) Number of states in a sequential circuit

B) Number of outputs

C) Clock frequency

D) Number of inputs

Correct answer: A) Number of states in a sequential circuit
Equivalent states are merged to reduce circuit complexity.

12. Edge-triggered flip-flops respond to:

A) Entire clock pulse

B) Clock level

C) Clock transition

D) Data input only

Correct answer: C) Clock transition
They sample input at a clock edge.

13. In an asynchronous counter, the clock input is applied to:

A) All flip-flops simultaneously

B) First flip-flop only

C) Last flip-flop only

D) No flip-flop

Correct answer: B) First flip-flop only
The first flip-flop receives the external clock; later stages are triggered by preceding stages.

14. Which statement is true regarding synchronous counters?

A) Each flip-flop receives a different clock

B) All flip-flops are triggered simultaneously

C) Propagation delay accumulates stage by stage

D) Only asynchronous reset is possible

Correct answer: B) All flip-flops are triggered simultaneously
All flip-flops share the same clock signal.

15. A register is primarily used to:

A) Generate clock pulses

B) Perform arithmetic operations

C) Store binary information temporarily

D) Decode instructions

Correct answer: C) Store binary information temporarily
Registers provide temporary storage for binary data.

16. State minimization is carried out to:

A) Reduce equivalent states

B) Increase flip-flops

C) Increase clock speed

D) Reduce inputs

Correct answer: A) Reduce equivalent states
Equivalent states are combined to simplify the state machine.

17. In a Moore machine, the output depends on:

A) Present state only

B) Input only

C) Present state and input

D) Next state

Correct answer: A) Present state only
Moore outputs depend only on the present state.

18. The number of flip-flops required to construct an 8-bit register is:

A) 4

B) 8

C) 16

D) 32

Correct answer: B) 8
One flip-flop stores one bit.

19. Two states are equivalent if they:

A) Have the same binary assignment

B) Produce identical outputs and next states for the same inputs

C) Have different outputs

D) Use the same flip-flop

Correct answer: B) Produce identical outputs and next states for the same inputs
Equivalent states have matching outputs and equivalent transitions for every input.

20. State assignment refers to:

A) Assigning binary codes to states

B) Reducing states

C) Counting states

D) Removing transitions

Correct answer: A) Assigning binary codes to states
State assignment gives each state a binary representation.

UNIT III – COMPUTER FUNDAMENTALS

PART A – 1 MARK QUESTIONS

1. A computer system that uses transistors instead of vacuum tubes belongs to which generation?

A) First

B) Second

C) Third

D) Fourth

Correct answer: B) Second
Second-generation computers used transistors.

2. Which advancement significantly reduced computer size during the third generation?

A) Vacuum tubes

B) Integrated circuits

C) Magnetic drums

D) Relays

Correct answer: B) Integrated circuits
Integrated circuits combined many components on a small chip.

3. In a Von Neumann computer, instructions and data are stored in:

A) Separate memories

B) Main memory

C) Cache only

D) Registers only

Correct answer: B) Main memory
The stored-program Von Neumann model uses a common main memory for instructions and data.

4. Which register typically contains the address of the next instruction?

A) PC

B) IR

C) MDR

D) MAR

Correct answer: A) PC
The Program Counter holds the address of the next instruction.

5. Which component executes arithmetic and logical operations?

A) ALU

B) Control Unit

C) Bus

D) Memory Unit

Correct answer: A) ALU
The ALU performs arithmetic and logical operations.

6. Storing data from register R1 into memory location 4500 primarily uses:

A) Direct addressing

B) Immediate addressing

C) Relative addressing

D) Register addressing

Correct answer: A) Direct addressing
The memory address is explicitly specified.

7. Which addressing mode is most suitable for implementing arrays?

A) Immediate

B) Indexed

C) Implied

D) Direct

Correct answer: B) Indexed
Indexed addressing combines a base address and index/offset.

8. During instruction execution, the operand address is stored in:

A) MAR

B) IR

C) MDR

D) PC

Correct answer: A) MAR
The Memory Address Register holds the address used for a memory access.

9. Which instruction sequence is most likely for A = B + C?

A) ADD B, C, A

B) MOVE A, B

C) LOAD C; ADD

D) LOAD B; LOAD C; ADD; STORE A

Correct answer: D) LOAD B; LOAD C; ADD; STORE A
The sequence loads operands, adds them, and stores the result.

10. MOV R1, (R2), where R2 contains the operand address, uses:

A) Register indirect

B) Indexed

C) Immediate

D) Direct

Correct answer: A) Register indirect
The register contains the address of the operand.

11. During instruction sequencing, the Program Counter is updated to:

A) Store data

B) Hold the next instruction address

C) Perform arithmetic

D) Transfer control signals

Correct answer: B) Hold the next instruction address
The PC tracks the next instruction address.

12. When a branch instruction is executed, the processor:

A) Continues sequential execution

B) Changes the next instruction address

C) Clears memory

D) Resets the CPU

Correct answer: B) Changes the next instruction address
A branch modifies the PC when its condition or target is applied.

13. A high-level language program is converted into machine code using a:

A) Loader

B) Assembler

C) Linker

D) Compiler

Correct answer: D) Compiler
A compiler translates high-level source code.

14. In LOAD R1, X, X represents:

A) Opcode

B) Operand address

C) Register

D) Instruction code

Correct answer: B) Operand address
X identifies the operand location.

15. Which instruction commonly transfers data from CPU to memory?

A) LOAD

B) MOVE

C) ADD

D) STORE

Correct answer: D) STORE
STORE writes data from a register/CPU to memory.

16. Data transfer between CPU and memory takes place through the:

A) Data bus

B) Address bus

C) Program Counter

D) Control bus

Correct answer: A) Data bus
The data bus carries the actual data.

17. Which bus carries read and write signals?

A) Data bus

B) Address bus

C) Control bus

D) System clock

Correct answer: C) Control bus
Read/write are control signals.

18. A unique identifier assigned to a memory location is called:

A) Operand

B) Opcode

C) Address

D) Register

Correct answer: C) Address
A memory address identifies a location.

19. Reducing CPI while keeping all other factors constant will:

A) Increase execution time

B) Improve performance

C) Reduce memory size

D) Decrease clock rate

Correct answer: B) Improve performance
Lower CPI reduces the number of clock cycles required.

20. ISA acts as an interface between:

A) Hardware and software

B) CPU and memory

C) Input and output

D) Cache and main memory

Correct answer: A) Hardware and software
The ISA defines the software-visible processor architecture.

UNIT IV – PROCESSOR DATAPATH AND CONTROL UNIT

PART A – 1 MARK QUESTIONS

1. The first stage of instruction execution is:

A) Decode

B) Execute

C) Store

D) Fetch

Correct answer: D) Fetch
The processor first fetches the instruction.

2. The datapath of a processor primarily performs:

A) Control signal generation

B) Data movement and processing

C) Interrupt handling

D) Cache mapping

Correct answer: B) Data movement and processing
The datapath contains registers, ALU and paths that process/move data.

3. Register Transfer Language (RTL) is used to describe:

A) Program execution

B) Data transfer between registers

C) I/O communication

D) Cache organization

Correct answer: B) Data transfer between registers
RTL describes micro-operations and transfers among registers.

4. Which register stores the address of the next instruction?

A) MAR

B) MDR

C) IP

D) PC

Correct answer: D) PC
The Program Counter stores the next instruction address.

5. A hardwired control unit is generally _____ than a microprogrammed control unit:

A) Slower

B) Faster

C) Cheaper

D) More flexible

Correct answer: B) Faster
Hardwired control generates signals directly through logic circuits.

6. Microprogrammed control generates control signals using:

A) Registers

B) Control memory

C) Cache

D) ALU

Correct answer: B) Control memory
Microinstructions stored in control memory generate control signals.

7. Instruction pipelining primarily improves processor:

A) Cost

B) Throughput

C) Power consumption

D) Memory size

Correct answer: B) Throughput
Pipelining overlaps instructions to increase completed instructions per unit time.

8. Which of the following is NOT a pipeline hazard?

A) Structural

B) Data

C) Control

D) Address

Correct answer: D) Address
The standard hazard classes are structural, data and control.

9. A data hazard occurs when:

A) Hardware resources conflict

B) An instruction depends on the previous instruction's result

C) A branch occurs

D) Memory is full

Correct answer: B) An instruction depends on the previous instruction's result
A dependency between instructions can delay execution.

10. ASIC stands for:

A) Advanced System IC

B) Application Specific Integrated Circuit

C) Application Standard IC

D) Automatic System IC

Correct answer: B) Application Specific Integrated Circuit
ASIC means Application-Specific Integrated Circuit.

11. A processor requiring frequent instruction updates should preferably use:

A) Structural

B) Data

C) Control

D) Reconfigurable implementation

Correct answer: D) Reconfigurable implementation
A reconfigurable platform such as FPGA supports updates after fabrication.

12. Operand forwarding is mainly used to minimize:

A) Structural hazards

B) Data hazards

C) Control hazards

D) Memory hazards

Correct answer: B) Data hazards
Forwarding supplies a result directly to a dependent instruction.

13. A branch instruction causing pipeline delay results in a _____ hazard:

A) Structural

B) Data

C) Resource

D) Control

Correct answer: D) Control
Branches create control hazards.

14. Increasing pipeline stages mainly aims to improve:

A) Cache size

B) Throughput

C) Memory capacity

D) Instruction size

Correct answer: B) Throughput
More stages can raise throughput, subject to overhead and hazards.

15. Which architecture is most suitable for highly pipelined processors?

A) CISC

B) ASIC

C) FPGA

D) RISC

Correct answer: D) RISC
RISC designs commonly use simpler, regular instructions that suit pipelining.

16. Structural hazards can be minimized by:

A) Operand forwarding

B) Hardware duplication

C) DMA

D) Polling

Correct answer: B) Hardware duplication
Duplicating or separating resources removes resource conflicts.

17. When is FPGA preferred over ASIC?

A) Reconfigurability is required

B) Lowest power is required

C) Mass production only

D) Highest speed is required

Correct answer: A) Reconfigurability is required
FPGA can be reprogrammed after manufacture.

18. Pipeline stalls mainly reduce processor:

A) Throughput

B) Register size

C) Instruction length

D) Memory capacity

Correct answer: A) Throughput
Stalls insert idle cycles and reduce instruction throughput.

19. Compared with FPGA, ASIC generally provides:

A) Better flexibility

B) Better performance

C) Easier modification

D) Lower design effort

Correct answer: B) Better performance
An application-specific design can be optimized for speed and power.

20. Branch prediction is primarily used to improve:

A) Cache performance

B) Pipeline efficiency

C) Memory bandwidth

D) Register transfer

Correct answer: B) Pipeline efficiency
Predicting branch outcomes reduces control-hazard penalties.

UNIT V – MEMORY AND I/O ORGANIZATION

PART A – 1 MARK QUESTIONS

1. Which memory is volatile and loses its contents when power is switched off?

A) ROM

B) Flash memory

C) Hard disk

D) SRAM

Correct answer: D) SRAM
SRAM is volatile memory.

2. Which memory is commonly used for main memory because of its high storage density?

A) ROM

B) SRAM

C) Cache memory

D) DRAM

Correct answer: D) DRAM
DRAM offers high density and is commonly used as main memory.

3. The main purpose of cache memory is to:

A) Increase storage capacity

B) Reduce memory access time

C) Replace secondary storage

D) Store permanent data

Correct answer: B) Reduce memory access time
Cache stores frequently used data close to the CPU.

4. Which cache mapping allows a memory block to be placed in any cache line?

A) Direct mapping

B) Set-associative mapping

C) Block mapping

D) Fully associative mapping

Correct answer: D) Fully associative mapping
Fully associative mapping permits placement in any cache line.

5. Which page replacement algorithm replaces the page not used for the longest time?

A) Optimal

B) FIFO

C) LRU

D) Random

Correct answer: C) LRU
LRU replaces the least recently used page.

6. Virtual memory enables a computer to:

A) Increase CPU speed

B) Execute programs larger than physical memory

C) Replace cache memory

D) Increase clock frequency

Correct answer: B) Execute programs larger than physical memory
Virtual memory provides a larger logical address space than available RAM.

7. Which storage device provides the highest access speed?

A) Magnetic tape

B) Hard disk

C) Optical disk

D) Cache memory

Correct answer: D) Cache memory
Cache is much faster than secondary storage.

8. Which I/O technique allows data transfer without continuous CPU intervention?

A) Polling

B) Interrupt-driven I/O

C) DMA

D) Program-controlled I/O

Correct answer: C) DMA
DMA transfers blocks between I/O and memory with limited CPU involvement.

9. A processor has a direct-mapped cache with 64 lines. A memory block can be mapped to:

A) Any one cache line

B) Only one specific cache line

C) Two cache lines

D) Four cache lines

Correct answer: B) Only one specific cache line
Direct mapping gives each memory block exactly one possible cache line.

10. A CPU repeatedly accesses the same data. Which organization gives the greatest performance improvement?

A) Larger hard disk

B) Cache memory

C) Optical disk

D) ROM

Correct answer: B) Cache memory
Cache exploits temporal and spatial locality.

11. During DMA transfer, who controls the system bus while data is transferred?

A) CPU

B) ALU

C) Interrupt controller

D) DMA controller

Correct answer: D) DMA controller
The DMA controller becomes bus master for the transfer.

12. A page fault occurs when:

A) Cache becomes full

B) Required page is not present in main memory

C) CPU executes an invalid instruction

D) DMA transfer fails

Correct answer: B) Required page is not present in main memory
A page fault signals that the requested virtual-memory page is not resident in RAM.

13. Which page replacement algorithm generally performs well when recently accessed pages are likely to be reused?

A) FIFO

B) LRU

C) Random

D) MRU

Correct answer: B) LRU
LRU uses recent access history as a predictor of reuse.

14. Which standard interface connects peripherals such as keyboards, printers and flash drives?

A) PCI

B) USB

C) AGP

D) HDMI

Correct answer: B) USB
USB is a common peripheral interface.

15. Which interface standard connects high-speed expansion cards to the motherboard?

A) USB

B) PCI

C) SATA

D) HDMI

Correct answer: B) PCI
PCI is an expansion-bus interface.

16. Which page replacement algorithm replaces the page that entered memory first?

A) FIFO

B) LRU

C) Optimal

D) LFU

Correct answer: A) FIFO
FIFO removes the oldest loaded page.

17. Which interrupt is generated by external I/O devices requesting CPU service?

A) Software interrupt

B) Internal interrupt

C) Hardware interrupt

D) Trap

Correct answer: C) Hardware interrupt
External devices generate hardware interrupts.

18. Which interface allows multiple peripheral devices on a bus using unique device IDs?

A) USB

B) PCI

C) SCSI

D) HDMI

Correct answer: C) SCSI
SCSI supports multiple addressed devices on a shared bus.

19. SRAM differs from DRAM because SRAM:

A) Requires periodic refreshing

B) Has higher storage density

C) Does not require refreshing

D) Stores data permanently

Correct answer: C) Does not require refreshing
SRAM stores bits in bistable circuits and does not need periodic refresh.

20. Which ROM can be erased by exposing it to ultraviolet light?

A) PROM

B) EPROM

C) Flash ROM

D) EEPROM

Correct answer: B) EPROM
EPROM is erased using ultraviolet light.

QUICK ANSWER KEY

UNIT I – COMBINATIONAL LOGIC

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UNIT V – MEMORY AND I/O ORGANIZATION

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